

1. (16 points) Prove that every function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ can be computed by a circuit of size less than $10 \cdot 2^n$.
2. (21 points) Improve this bound to show that every function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ can be computed by a circuit of size less than $1000 \cdot 2^n/n$.
3. (21 points) Show that for every $k > 0$ that PH contains languages whose circuit complexity is $\Omega(n^k)$.
4. (21 points) Show that if $P = NP$, then there is a language in EXP that requires circuits of size $2^n/n$.
5. (21 points) Show that $\text{uniform NC}^1 \subseteq L$. Then show that $\text{PSPACE} \neq \text{uniform NC}^1$.